

**Sternoclavicular · DoF1 — Protraction(+)/retraction(-) — sagittal plane elevation · worst-fitted of the 16 shoulders K was chosen on (all rows (ex + in vivo)), subject-specific prediction, K = 4**

**per-shoulder fit**  
worst/median =  $1.43 \times 10^{-1}$  of 16 flagged (robust z > 3)

